Université de la Science et de la Technologie HOUARI BOUMEDIENE

StepZ

Tapis piézoélectrique

Benchikh Le Houcine Chaima

M1 PHYSIQUE DES MATERIAUX

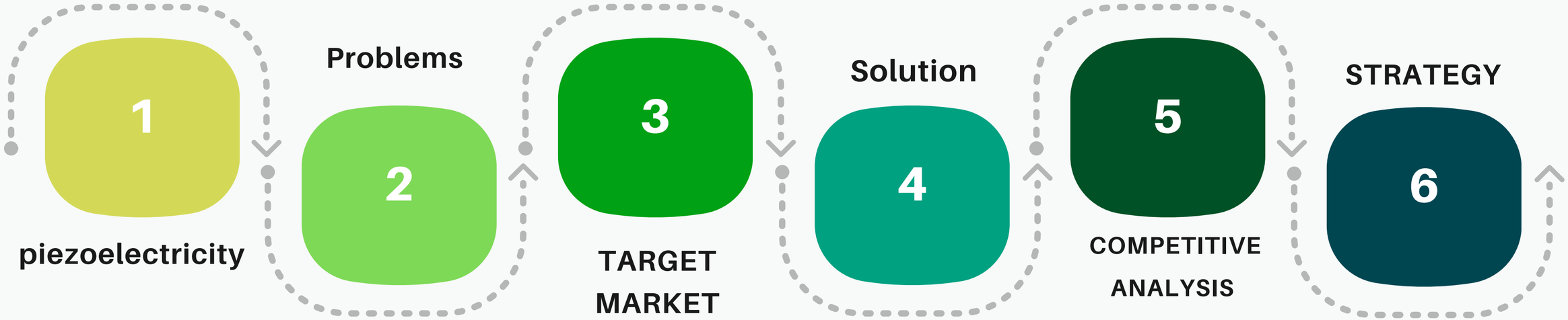
2024_2025

Introduction :

“THE Law of Conservation of Energy states that Energy can neither be created nor destroyed, rather, it can only be transformed or transferred from one form to another. this project explores the notation of exploitation of the kinetic energy released at the pressure of footsteps exerted on the carpet .



PLAN

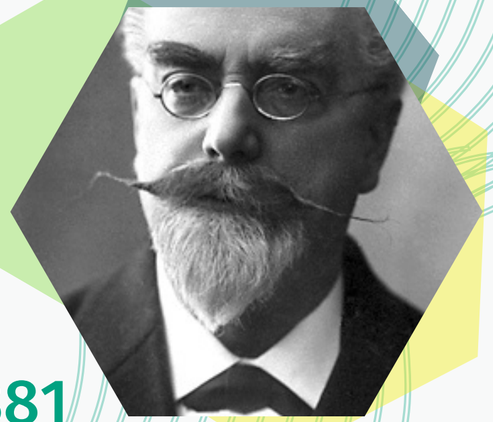

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History



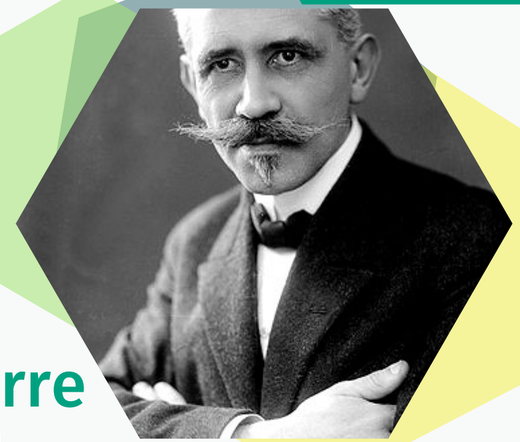
1880

Pierre and Jacques Curie discovered the piezoelectric effect.



1881

Gabriel Lippmannthe inverse piezoelectric effect.

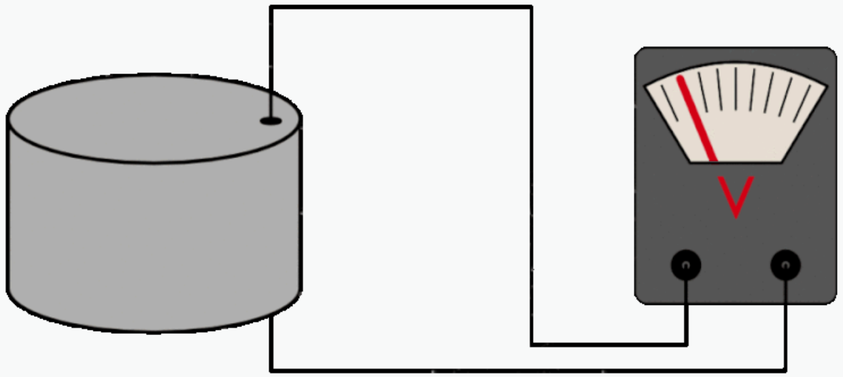


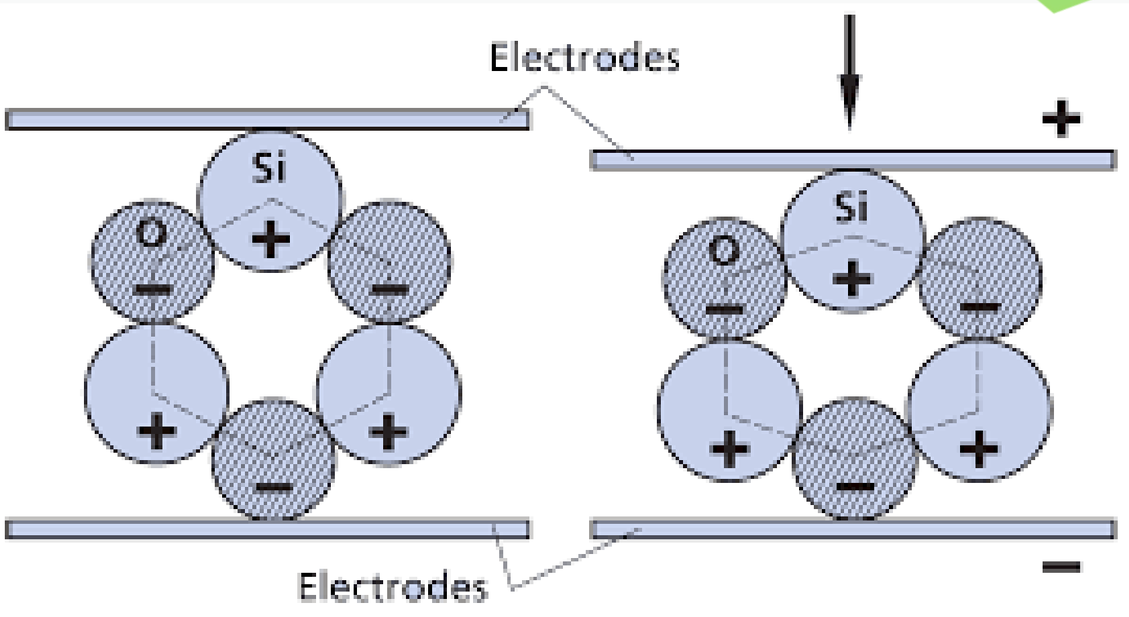
1 er Guerre mondiale

The first practical application of piezoelectricity was sonar, developed by Paul Langevin.

Piezoelectricity:

Piezo Material





Problems :



pollution



maintenance



Cost

TARGET MARKET :

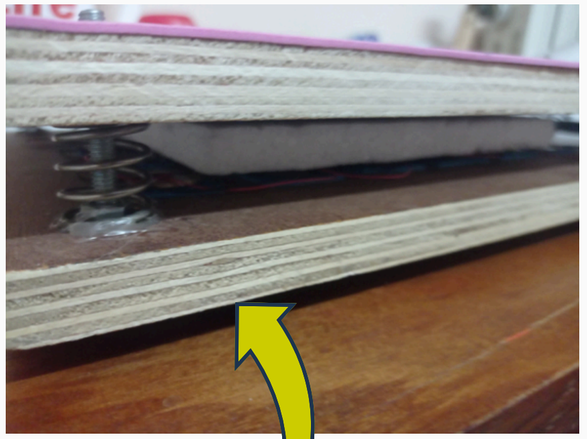
The target market for a piezoelectric walkway project in Algeria would primarily be:

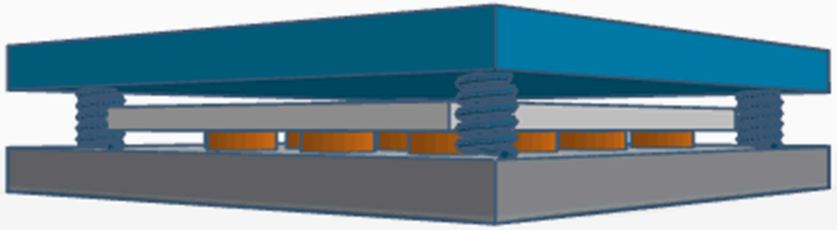
1. ***Government Entities (BtoG):*** As the government is aiming for a 27% renewable share in electricity generation by 2030, they would be a key target. The piezoelectric walkway could be part of the solution to achieve this target.

2.Businesses (BtoB): Businesses with high foot traffic, such as shopping malls, could be interested in this solution as a way to reduce their energy costs and promote their commitment to sustainability.

Solution:

- **Clean and sustainable**
- **Cost-effective :** low cost per unit of energy generated
- **highly reliable** and can continue to generate electricity
- highly **Efficient**
- **Versatile**
- **Low maintenance**





COMPETITION ANALYSIS :

Indirect Competitors:

SHAEMS
شمس
الشركة الجزائرية للطاقات المتجددة
ALGERIAN RENEWABLE ENERGY COMPANY

سوناطراك
sonatrach

SONELGAZ

BELECTRIC®

AEC
Algerian Energy Company Spa

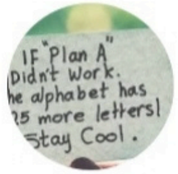
AFD
AGENCE FRANÇAISE
DE DÉVELOPPEMENT

ABM
Building Value

Mouloud Mebarki mar

Bonjour, veuillez trouver, en fichier attaché, deux document qui peuvent t'aider à comprendre mieux les matériaux piézoélectriques.Bon courage.

شكرا Merci



sm USTHB 09/04/2021

quelques liens: https://www.researchgate
.net/publication/235987254_Generating

Thanks

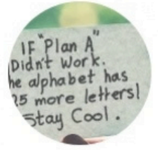
amar laib

Bonsoir, Demain inchallah je serai au labo de 10h00 à 11h00. \



Yacine SALHI

bonsoir, Vous pouvez commencer dès aujourd'hui mercredi



sm USTHB 1 Jun

Je suis heureux de cet exploit,

RABBI ywaffaq!